

Data Analytics

Micro Projects - Topic 3: Public Transport

Introduction:

You and your team are part of a leading data science company that specializes in transportation analytics. One of your major clients, a municipal transportation authority, is facing challenges with public transport delays that disrupt daily commuting and reduce system efficiency. Your task is to develop a predictive model to forecast arrival and departure delays for each station and trip in the public transportation network. The model should accurately predict the delay duration in minutes for both arrival and departure at each station listed in the dataset. The dataset provided contains detailed records of transit operations, including scheduled and actual times for arrivals and departures at various stations.

Tasks:

- Data Understanding and Cleanup: Describe your dataset and give an insight into the data you're working with. This should also include visualization and giving an insight into its most important features.
- Choosing a model: Choose a model for your specific problem. This should include a definition of the prediction problem. Compare different models and explain which model you have chosen and why it's the best suited one for your project.
- Evaluation: Talk about the experiences you have collected during your project. What were your biggest challenges, how did you overcome them and talk about your biggest lesson learned during the project. evaluate the trained model on unseen data.
- Creating a presentation: Your presentation should include and represent the most important aspect of the tasks named above. It should also include a brief description of the single steps that you've taken to achieve your goals.

Bonus Points:

Creative solutions such as feature engineering, the collection of additional data, or novel visualization techniques are encouraged and will positively impact your bonus points.